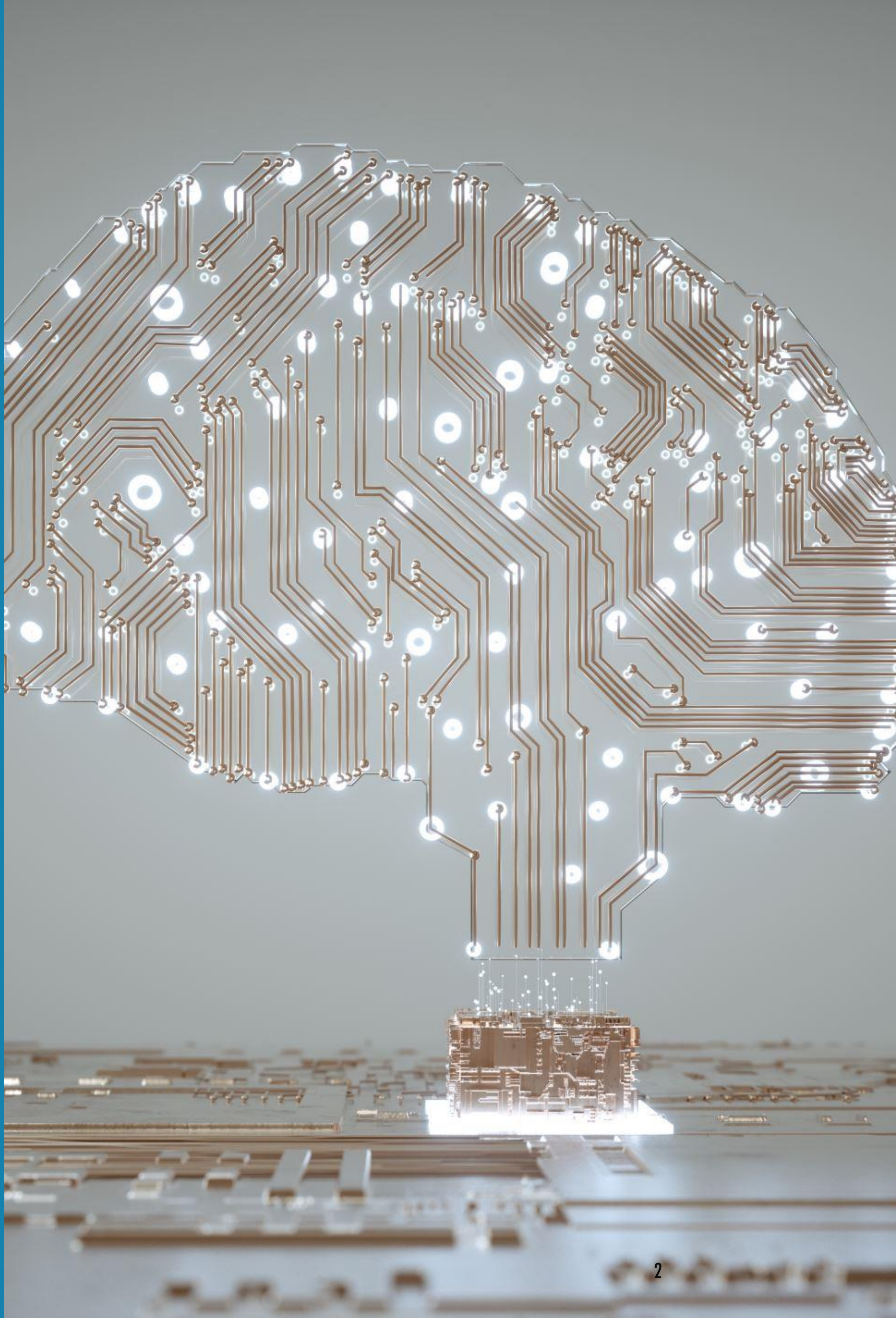
DEEP LEARNING

An Overview of Deep Learning, Types, Use Cases, and Applications

INTRODUCTION TO DEEP LEARNING

Deep Learning is a subset of machine learning in artificial intelligence (AI) that has networks capable of learning unsupervised from data that is unstructured or unlabeled. Also known as deep neural learning or deep neural network.

It differs from traditional machine learning by its capability to process a large amount of features and learn complex patterns in data.



VIDEO

https://www.youtube.com/watch?v=6M5VXKLf4D4&t=210s

https://www.youtube.com/watch?v=rEDzUT3ymw4

HISTORY

https://sefiks.com/2017/10/14/evolution-of-neural-networks/

https://sds-aau.github.io/SDS-master/M3/notebooks/ANN_intro.html#1

VOCABULARY IN DEEP LEARNING

1. Neurons: Basic units of a neural network that receive input, process it, and pass it on to the next layer.

2. Layers: Different levels in a neural network where computations are performed.

3. Activation Functions: Functions that determine whether a neuron should be activated or not.

4. Loss Function: A function that measures how well the neural network is performing.

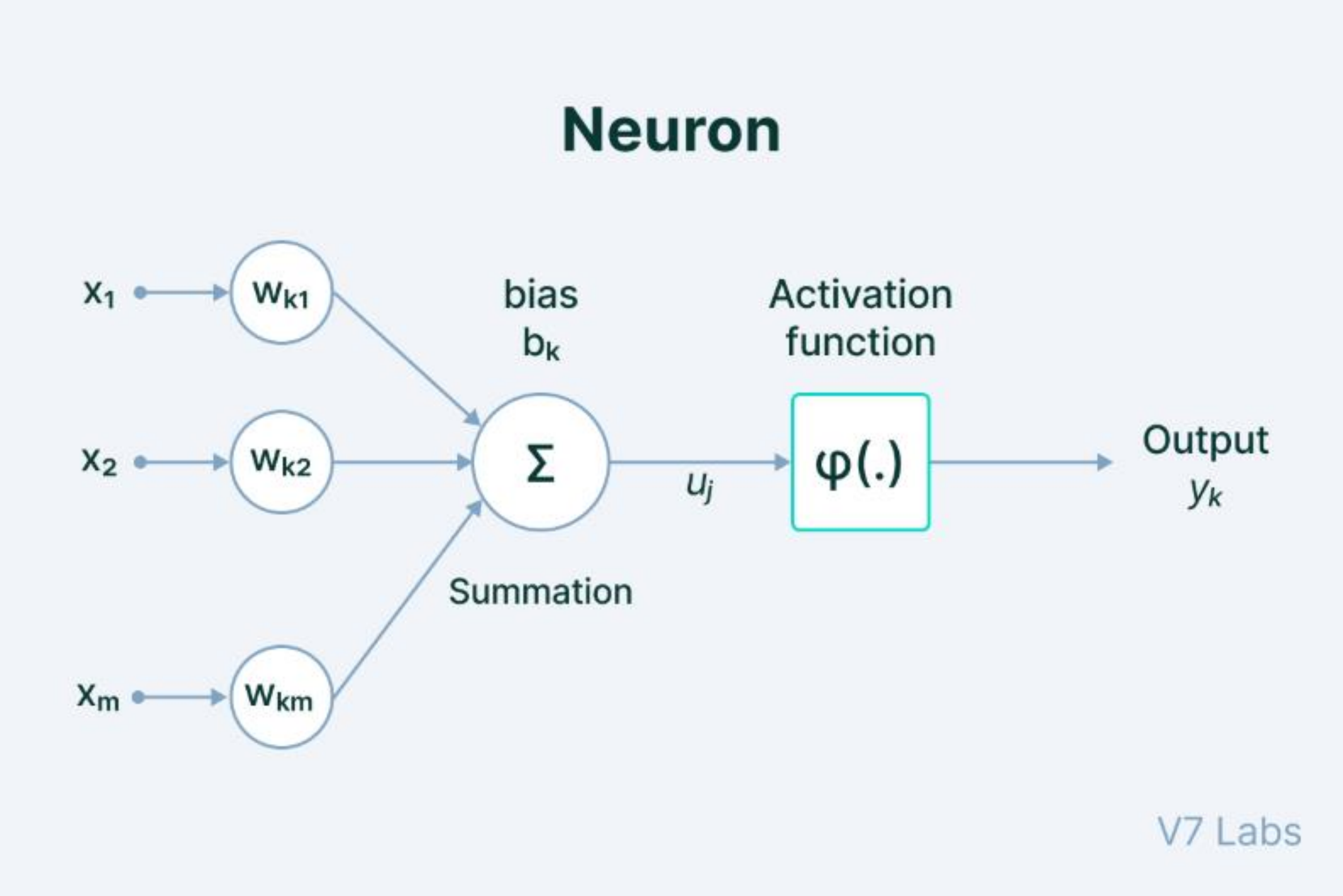
5. Optimizer: An algorithm used to minimize the loss function.

6. Epoch: One complete pass through the entire training dataset.

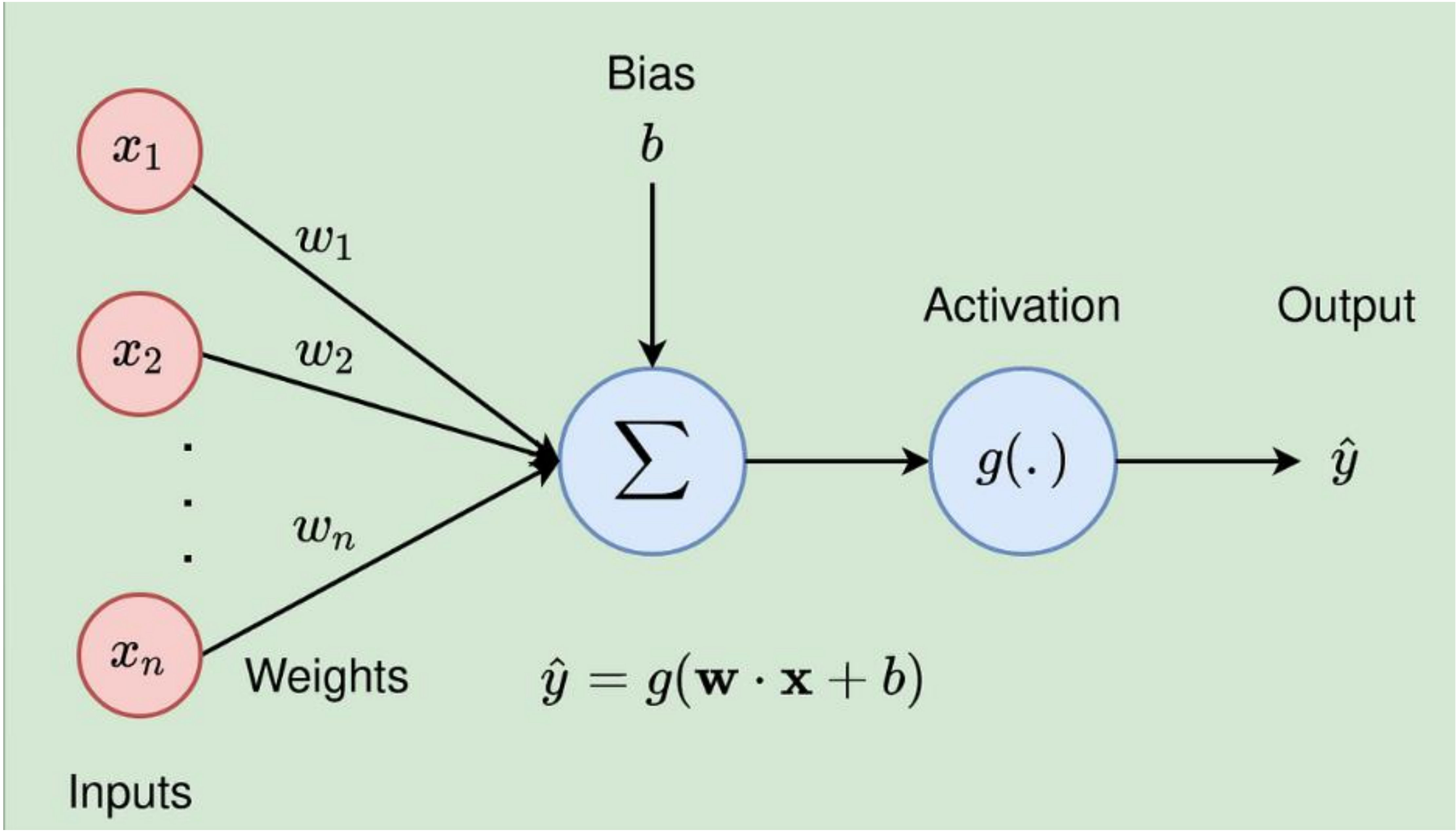
NEURAL NETWORKS

Neural Networks are a subset of machine learning and are at the heart of deep learning algorithms. Their name and structure are inspired by the human brain, mimicking the way that biological neurons signal to one another.

A neural network consists of layers of nodes, with each layer connected to the next. Each node is a perceptron, similar to a multiple linear regression.



https://assets-global.website-files.com/5d7b77b063a9066d83e1209c/614fc05e2486109794ed3bdc_neuron.png


Bias
b
x_1
w_1
x_2
w_2
Σ
Activation
Output
$g(.)$
$\hat{y}$
w_n
x_n
Weights
$\hat{y} = g(\mathbf{w} \cdot \mathbf{x} + b)$
Inputs

CORE CONCEPTS IN DEEP LEARNING

1. Neural Networks: A series of algorithms that attempt to recognize underlying relationships in a set of data through a process that mimics the way the human brain operates.

2. Forward Propagation: The process by which input data is passed through the neural network to generate an output.

3. Backward Propagation: The process by which the neural network learns from its mistakes by adjusting weights based on the error of the output.

4. Feature Engineering: The process of using domain knowledge to extract features from raw data that make machine learning algorithms work.

ARTICLE REVIEW

https://www.geeksforgeeks.org/neural-networks-a-beginners-guide/

DEEP LEARNING ALGORITHMS

1. Convolutional Neural Networks (CNNs): Primarily used for image and video recognition. They are designed to automatically and adaptively learn spatial hierarchies of features from input images.

2. Recurrent Neural Networks (RNNs): Used for sequential data and time series analysis. They have connections that form directed cycles, allowing information to persist.

3. Generative Adversarial Networks (GANs): Used for generating new data that resembles the training data. They consist of two neural networks, one generating candidates (the generator) and the other evaluating them (the discriminator).

4. Transformers: Used for natural language processing tasks. They rely on a mechanism called attention to draw global dependencies between input and output.

STRUCTURE OF A NEURAL NETWORK

1. Neurons: Basic units of a neural network that receive input, process it, and pass it on to the next layer.

2. Layers: Different levels in a neural network where computations are performed.

   - Input Layer: The first layer that receives the input data.

   - Hidden Layers: Intermediate layers that perform computations and extract features.

   - Output Layer: The final layer that produces the output.

WORKING OF A BASIC NEURAL NETWORK

1. Forward Propagation: The process by which input data is passed through the neural network to generate an output.

- Each neuron receives input from the previous layer, applies a weight to it, adds a bias, and passes it through an activation function.

- The output of each neuron becomes the input for the next layer.

2. Activation Functions: Functions that determine whether a neuron should be activated or not.

3. Backward Propagation: The process by which the neural network learns from its mistakes by adjusting weights based on the error of the output.

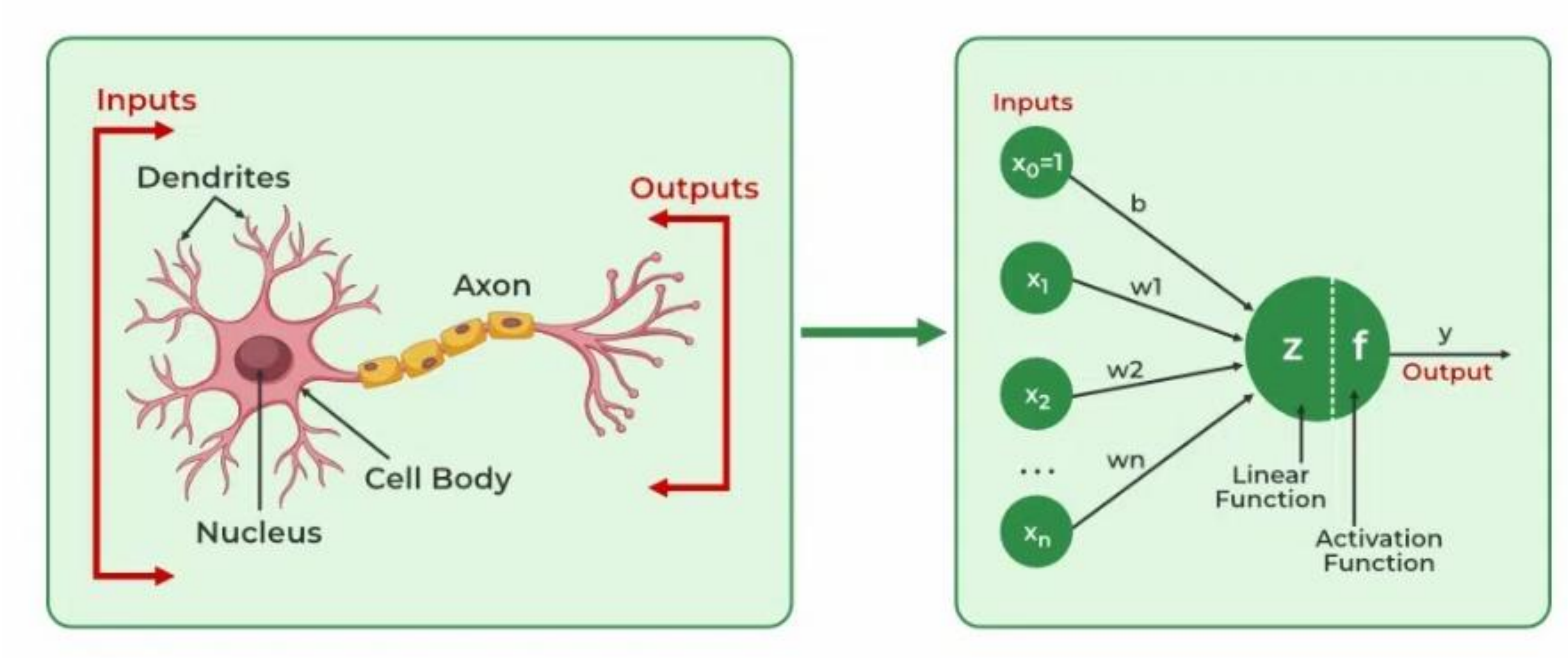
- The error is calculated using a loss function.

- The gradients of the loss function with respect to each weight are calculated and used to update the weights.

LEARNING IN NEURAL NETWORKS

Learning in neural networks follows a structured, three-stage process:

1. **Input Computation**: Data is fed into the network.
2. **Output Generation**: Based on the current parameters, the network generates an output.
3. **Iterative Refinement**: The network refines its output by adjusting weights and biases, gradually improving its performance on diverse tasks.



Img src: https://www.geeksforgeeks.org/neural-networks-a-beginners-guide/

EXAMPLE OF A BASIC NEURAL NETWORK

Consider a simple neural network with one hidden layer:

1. Input Layer: Receives input data (e.g., pixel values of an image).

2. Hidden Layer: Applies weights, biases, and activation functions to extract features.

3. Output Layer: Produces the final output (e.g., class label for image classification).

Step-by-step process:

- Input data is passed through the input layer.
- The hidden layer processes the input data and extracts features.
- The output layer produces the final output based on the extracted features.

USE CASES OF DEEP LEARNING

1. Image and Speech Recognition: Used in applications like facial recognition, voice assistants, etc.

2. Natural Language Processing: Used in language translation, sentiment analysis, etc.

3. Autonomous Vehicles: Used in self-driving cars for object detection and decision making.

4. Healthcare Diagnostics: Used for medical image analysis, predicting diseases, etc.

APPLICATIONS OF DEEP LEARNING

1. Self-Driving Cars: Use deep learning for object detection, lane detection, etc.

2. Virtual Assistants: Use deep learning for speech recognition and natural language understanding.

3. Fraud Detection: Use deep learning to detect fraudulent activities in finance.

4. Personalized Recommendations: Use deep learning to provide personalized content recommendations.

CASE STUDIES

https://research.aimultiple.com/deep-learning-applications/

https://expertbeacon.com/deep-learning-applications/

VIDEO-MUST WATCH

https://www.youtube.com/watch?v=FbxTVRfQFuI&t=2233s

GLOSSARY-MUST READ

https://www.analyticsvidhya.com/blog/2017/05/25-must-know-terms-concepts-for-beginners-in-deep-learning/